

Store at: +2+8°C.

Presentation:
Cod. SU048 CONT: R1 5x8 + R2 1x10 mL

Procedure

Quantitative determination of zinc.

Only for in vitro use in clinical laboratory (IVD)

TEST SUMMARY

Direct colorimetric test without deproteinization of the sample. End point increase. At pH 8.2, in a buffered media, zinc reacts with the specific complexant NITRO-PAPS, forms a stable coloured complex. The colour intensity is proportional to the amount of zinc present in the sample.

REAGENTS COMPOSITION

R 1	Borate Buffer. pH 8.2	0.4 mol/L
Buffer		
R 2	NITRO-PAPS	0.4 mmol/L
Color		
ZINC CAL	Zinc primary standard 200 µg/dL	

PRECAUTIONS

R1: H318-Causes serious eye damage. H412- Harmful to aquatic life with long lasting effects.
Follow the precautionary statements given in MSDS and label of the product.

CALIBRATION

The zinc cal value is verified using NIST (National Institute of Standards and Technology) traceable reference standard.

REAGENT PREPARATION AND STABILITY

The reagents are ready to use. Reagents are stable until expiration date indicated on the label, if contamination avoided and vial recapped immediately after use and stored at 2-8°C.

ZINC CAL: Proceed carefully with this product because due its nature it can get contaminated easily.

Signs of reagent deterioration:

- Presence of particles and turbidity.

All the reagents of the kit are stable up to the end of the indicated month and year of expiry. Store tightly closed at 2-8°C. Do not use reagents over the expiration date.

SPECIMEN

Serum or plasma¹: Not hemolyzed. Use only heparin salts as anticoagulants. Stability: 7 days at 2-8°C.

Seminal fluid: Centrifuge the sample at 3000 r.p.m. for 10-15 minutes. Dilute supernatant 1/100 with NaCl 9 g/L and multiply the result by 100. Stability: 7 days at 2-8°C.

24 h Urine.

MATERIAL REQUIRED BUT NOT PROVIDED

- Spectrophotometer or colorimeter measuring at 578 nm.
- Matched cuvettes 1.0 cm. light path.

General laboratory equipment^(note 1)

TEST PROCEDURE

- Allow reagents to reach working temperature before using. A proportional variation of the reaction volumes indicated does not change the result.
- Assay conditions:
 - Wavelength:578 nm (550-580)
 - Cuvette: 1 cm light path
 - Temperature 25 / 30 / 37°C
- Adjust the instrument to zero with distilled water.
- Pipette into a cuvette:

	Blank	Standard	Sample
R1 (mL)	0.8	0.8	0.8
R2 (mL)	0.2	0.2	0.2
Distilled water	50	-	-
Standard ^(Note 2) (µL)	--	50	--
Sample (µL)	--	--	50

- Mix and incubate 5 minutes.
- Read the absorbance (A) of sample and standard against Blank. The colour is stable for at least 30 min.

CALCULATIONS

$$\frac{(A) \text{ Sample}}{(A) \text{ Standard}} \times 200 (\text{Standard conc.}) = \mu\text{g/dL Zinc in the sample}$$

Conversion factor: µg/dL x 0.153 = µmol/L.

QUALITY CONTROL

Control sera are recommended to monitor the performance of the procedure, Normal and Pathological.

If control values are found outside the defined range, check the instrument, reagents and calibrator for problems.

Serum controls are recommended for internal quality control. Each laboratory should

REFERENCE VALUES¹

Serum or plasma:

68 - 107 µg/dL $\approx$ 10.4 - 16.4 µmol/L

Centrifuged seminal fluid:

2 - 10 mg /dL $\approx$ (0.31 - 1.53 mmol/L)

24 h Urine:

150 - 1200 µmol/dL

(These values are for orientation purpose).

It is suggested that each laboratory establish its own reference range.

CLINICAL SIGNIFICANCE

Nutritional zinc deficiency in humans is fairly prevalent throughout the world, deficiency is characterized by growth retardation in children and adolescents, hypogonadism in males, mild dermatitis, poor appetite, delayed wound healing, abnormal dark adaptation, and mental lethargy and impaired immune responses.

Clinical diagnosis should not be made on a single test result; it should integrate clinical and other laboratory data.

REAGENT PERFORMANCE

- **Measuring Range:** From detection limit of 4 µg/dL. to linearity limit of 1000 µg/dL., under the described assay conditions.

If results obtained were greater than linearity limit, dilute the sample 1/10 with NaCl 9 g/L. and multiply result by 10.

- **Precision:**

	Intra-assay (n=20)			Inter-assay (n=20)		
Mean (µg/dL)	76.7	198	307	77.5	202	300
SD	2.29	4.60	3.38	1.14	0.90	3.27
CV (%)	2.95	2.27	1.13	1.46	0.45	1.09

- **Accuracy:** Results obtained GPL reagents (y) did not show systematic differences when compared with other commercial reagents (x).

The results obtained using 60 samples were the following:

Correlation coefficient (r)²: 0.99

Regression Equation: y= 1.027x + 2.7873

The results of the performance characteristics depend on the analyzer used.

INTERFERING SUBSTANCES

No interferences were observed to bilirubin up to 15 mg/dL, hemoglobin up to 0.5 g/dL and triglycerides up to 1000 mg/dL.

EDTA interfere in the test.

A list of drugs and other interfering substances with zinc determination has been reported by Young et. al^{3,4}.

NOTES

- It is recommended to use disposable material. If glassware is used the material should be scrupulously cleaned with hydrochloric acid 1 N and then thoroughly rinsed it with distilled water.
- Use clean disposable pipette tips for its dispensation.

BIBLIOGRAPHY

- Burtis A et al. Tietz Textbook of Clinical Chemistry, 3rd ed AACC 1999.
- Kaplan A et al. Clin Chem The C.V. Mosby Co.1984.
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